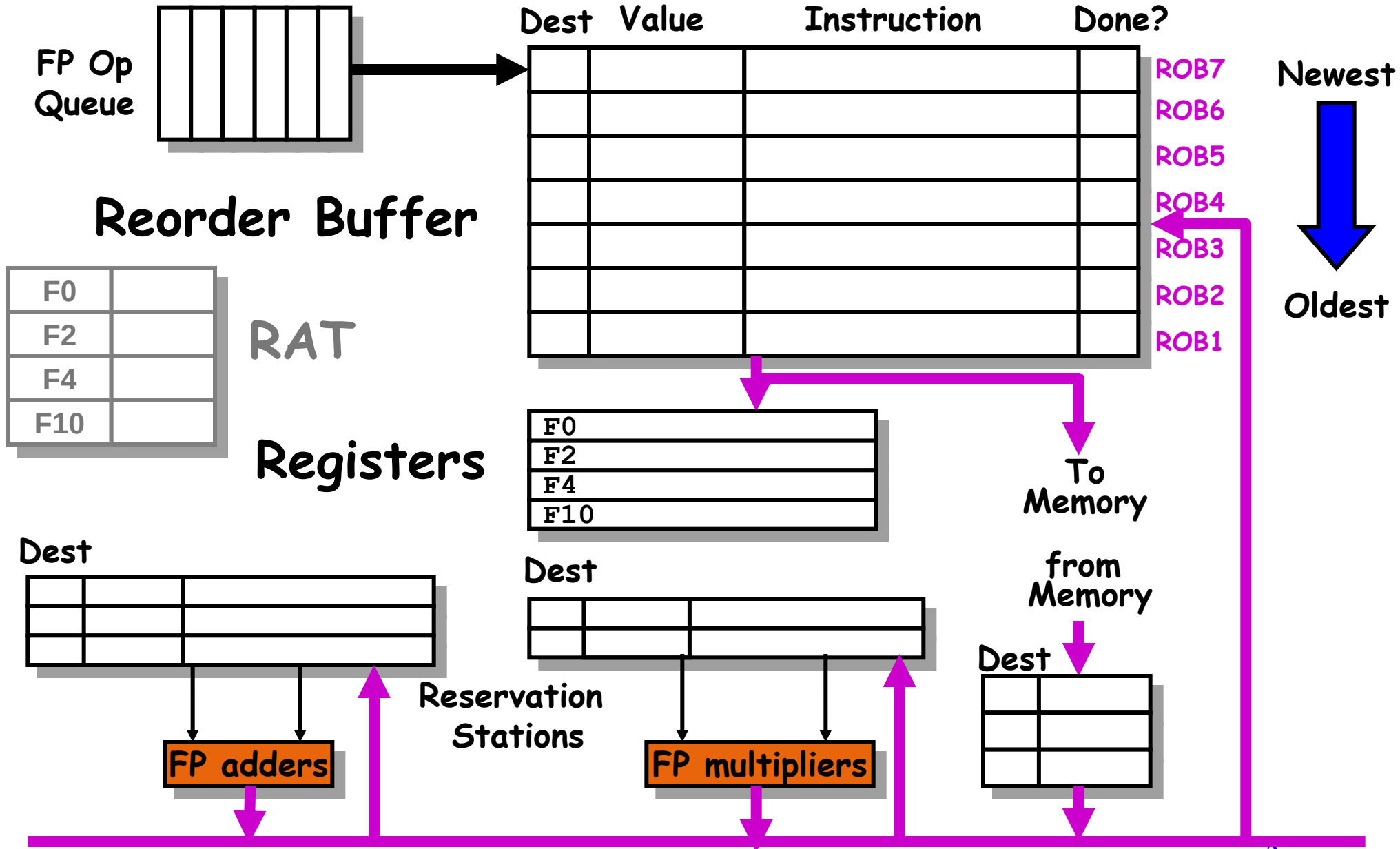


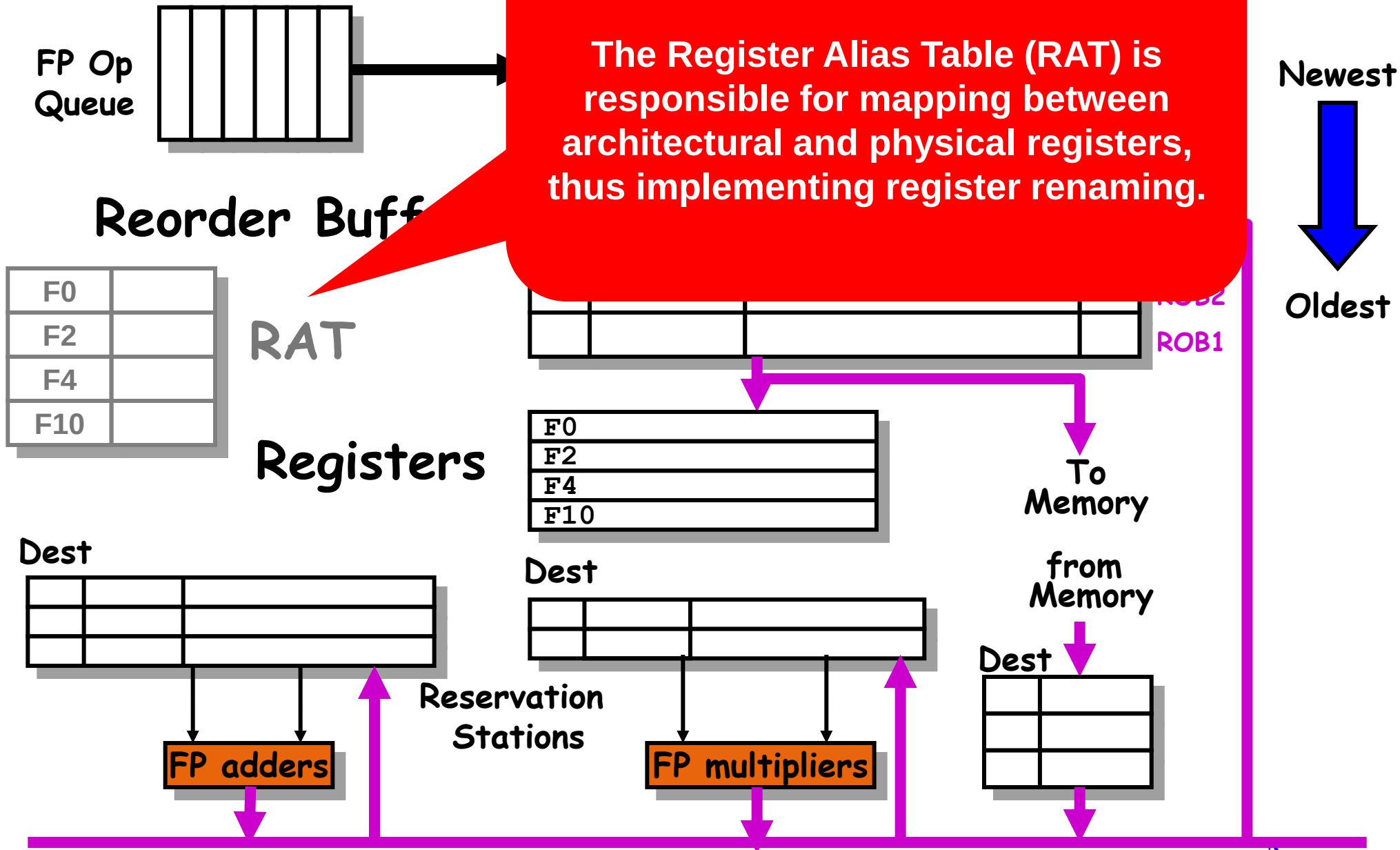
Example of code execution on a processor based on the Tomasulo architecture

*Slides modified from Prof. Dave Patterson,
Prof. John Kubiawicz, Prof. Nancy Warter-Perez (Berkeley Un.), and
Zvika Guz (Technion)*

Tomasulo With Reorder buffer: architecture



Tomasulo With Forwarding and Branch Prediction

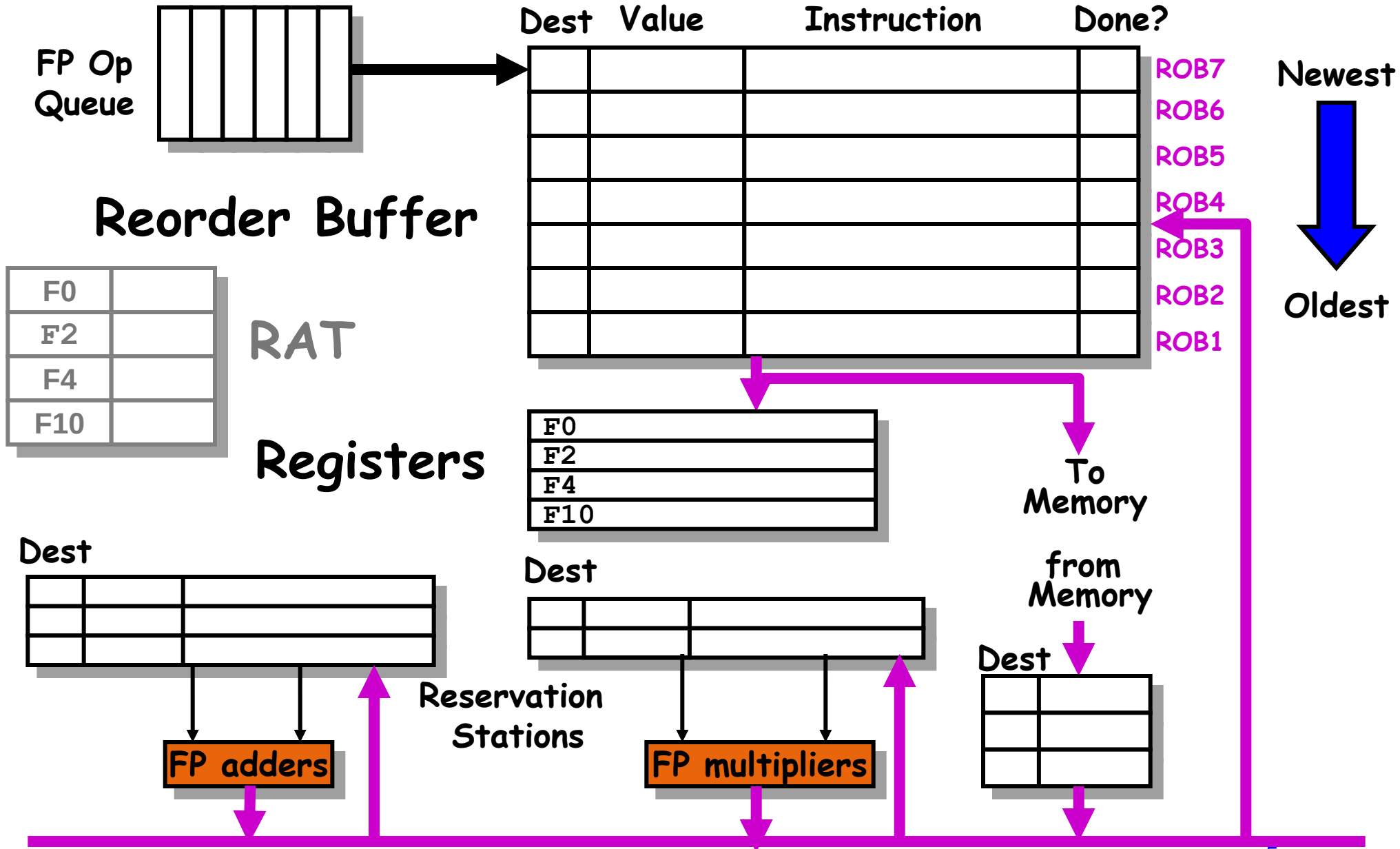


Code example

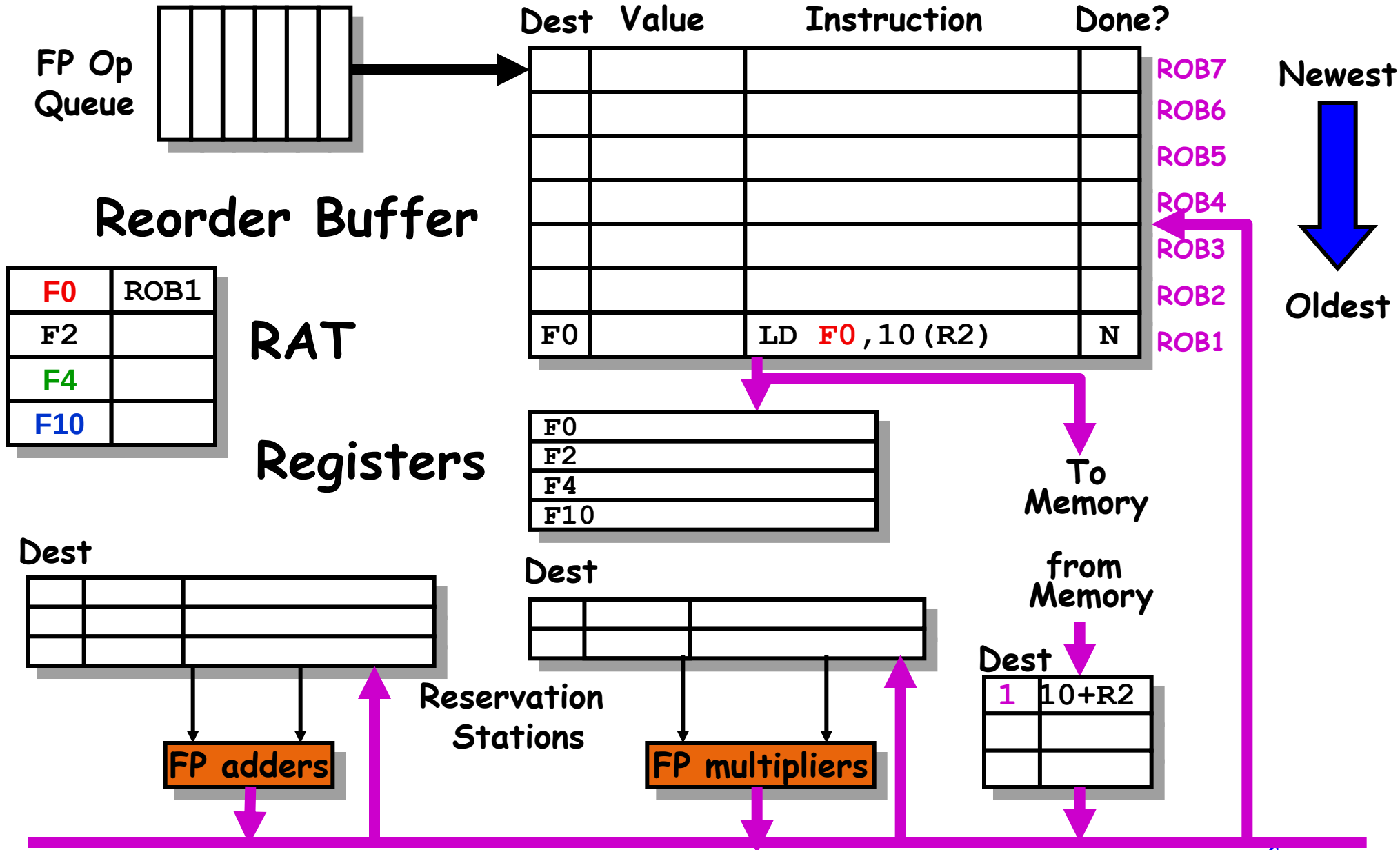
```
1. LD F0, 10(R2)
2. ADDD F10, F4, F0
3. DIVD F2, F10, F6
4. BNE F2, <LAB01>
5. LD F4, 0(R3)
6. ADDD F0, F4, F6
7. ADDD F0, F4, F6
```

<LAB01>

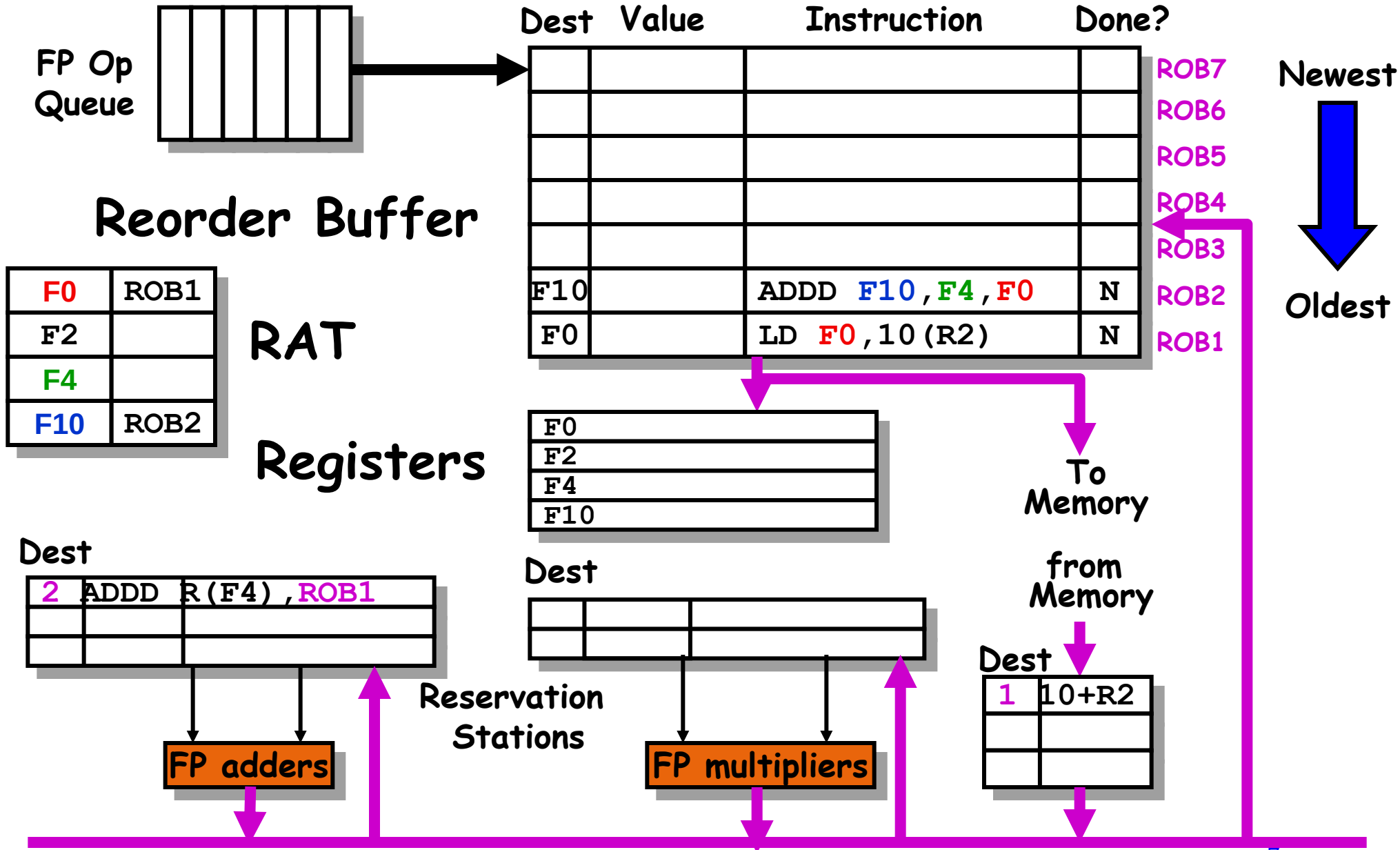
Tomasulo With Reorder buffer



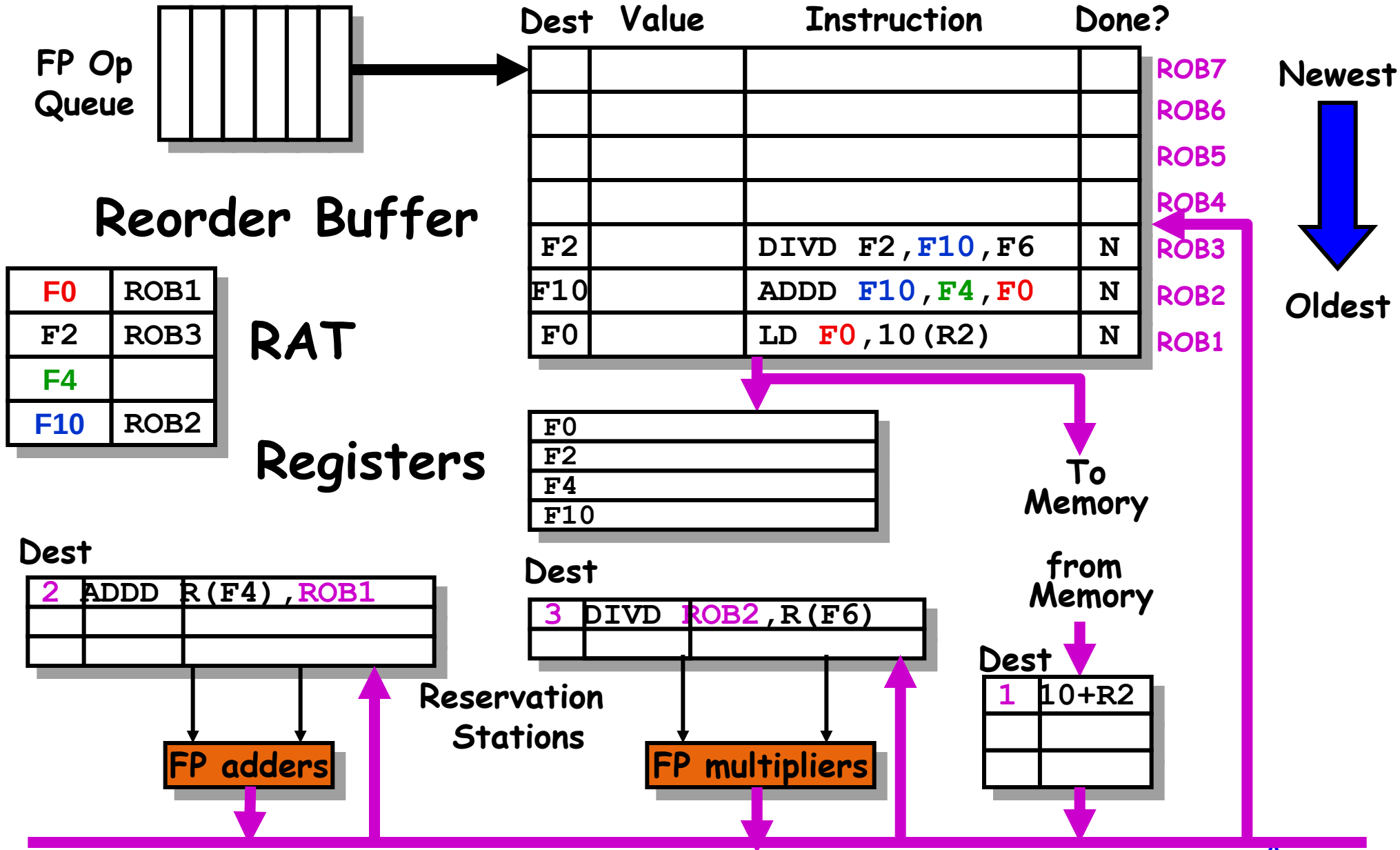
Tomasulo With Reorder buffer: 1



Tomasulo With Reorder buffer: 2

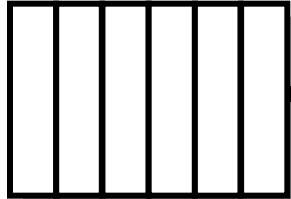


Tomasulo With Reorder buffer: 3



Tomasulo With Reorder buffer: 4, 5, 6

FP Op Queue



Reorder Buffer

F0	ROB6
F2	ROB3
F4	ROB5
F10	ROB2

RAT

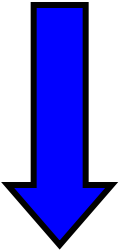
Registers

Dest Value Instruction Done?

F0		ADDD F0, F4, F6	N
F4		LD F4, 0 (R3)	N
--		BNE F2, <...>	N
F2		DIVD F2, F10, F6	N
F10		ADDD F10, F4, F0	N
F0		LD F0, 10 (R2)	N

ROB7
ROB6
ROB5
ROB4
ROB3
ROB2
ROB1

Newest



Oldest

F0
F2
F4
F10

To Memory

Dest

2	ADDD	R(F4), ROB1
6	ADDD	ROB5, R(F6)

FP adders

Reservation Stations

Dest

3	DIVD	ROB2, R(F6)

FP multipliers

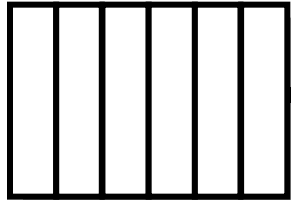
from Memory

Dest

1	10+R2
5	0+R3

Tomasulo With Reorder buffer: 7

FP Op Queue



Reorder Buffer

F0	ROB7
F2	ROB3
F4	ROB5
F10	ROB2

RAT

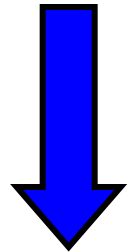
Registers

Dest Value Instruction Done?

F0		ADDD F0, F4, F6	N
F0		ADDD F0, F4, F6	N
F4		LD F4, 0 (R3)	N
--		BNE F2, <...>	N
F2		DIVD F2, F10, F6	N
F10		ADDD F10, F4, F0	N
F0		LD F0, 10 (R2)	N

ROB7
ROB6
ROB5
ROB4
ROB3
ROB2
ROB1

Newest



Oldest

F0
F2
F4
F10

To Memory

Dest

2	ADDD	R(F4), ROB1
6	ADDD	ROB5, R(F6)
7	ADDD	ROB5, R(F6)

FP adders

Dest

3	DIVD	ROB2, R(F6)
---	------	-------------

FP multipliers

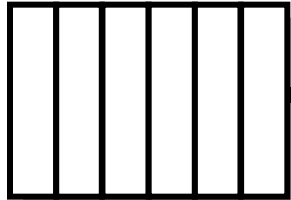
from Memory

Dest

1	10+R2
5	0+R3

Tomasulo With Reorder buffer: 8

FP Op Queue



Reorder Buffer

F0	ROB7
F2	ROB3
F4	ROB5
F10	ROB2

RAT

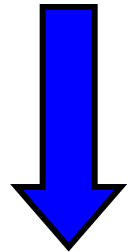
Registers

Dest Value Instruction Done?

F0		ADDD F0, F4, F6	N
F0		ADDD F0, F4, F6	N
F4	M[10]	LD F4, 0 (R3)	Y
--		BNE F2, <...>	N
F2		DIVD F2, F10, F6	N
F10		ADDD F10, F4, F0	N
F0		LD F0, 10 (R2)	N

ROB7
ROB6
ROB5
ROB4
ROB3
ROB2
ROB1

Newest



Oldest

F0
F2
F4
F10

To Memory

Dest

2	ADDD	R(F4), ROB1
6	ADDD	M[10], R(F6)
7	ADDD	M[10], R(F6)

FP adders

Reservation Stations

Dest

3	DIVD	ROB2, R(F6)

FP multipliers

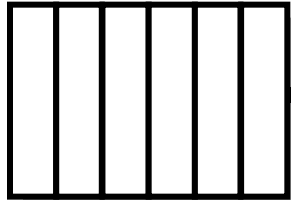
from Memory

Dest

1	10+R2

Tomasulo With Reorder buffer: 9

FP Op Queue



Reorder Buffer

F0	ROB7
F2	ROB3
F4	ROB5
F10	ROB2

RAT

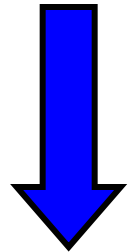
Registers

Dest Value Instruction Done?

F0	<val3>	ADDD F0, F4, F6	Y
F0	<val2>	ADDD F0, F4, F6	Y
F4	M[10]	LD F4, 0 (R3)	Y
--		BNE F2, <...>	N
F2		DIVD F2, F10, F6	N
F10		ADDD F10, F4, F0	N
F0		LD F0, 10 (R2)	N

ROB7
ROB6
ROB5
ROB4
ROB3
ROB2
ROB1

Newest



Oldest

F0
F2
F4
F10

To Memory

from Memory

Dest

2	ADDD R(F4), ROB1

Dest

3	DIVD ROB2, R(F6)

Dest

1	10+R2

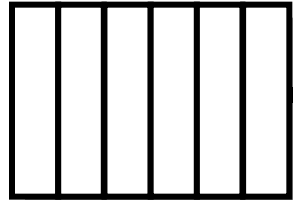
Reservation Stations

FP adders

FP multipliers

Tomasulo With Reorder buffer: 10

FP Op Queue



Reorder Buffer

F0	ROB7
F2	ROB3
F4	ROB5
F10	ROB2

RAT

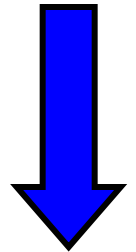
Registers

Dest Value Instruction Done?

F0	<val3>	ADDD F0, F4, F6	Y
F0	<val2>	ADDD F0, F4, F6	Y
F4	M[10]	LD F4, 0 (R3)	Y
--		BNE F2, <...>	N
F2		DIVD F2, F10, F6	N
F10		ADDD F10, F4, F0	N
F0	M[2]	LD F0, 10 (R2)	Y

ROB7
ROB6
ROB5
ROB4
ROB3
ROB2
ROB1

Newest



Oldest

F0
F2
F4
F10

To Memory

from Memory

Dest

2	ADDD R(F4), M[2]

Dest

3	DIVD ROB2, R(F6)

Dest

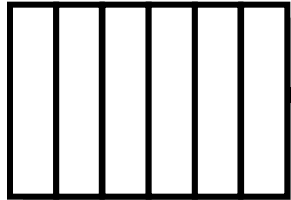
Reservation Stations

FP adders

FP multipliers

Tomasulo With Reorder buffer: 11

FP Op Queue



Reorder Buffer

F0	ROB7
F2	ROB3
F4	ROB5
F10	ROB2

RAT

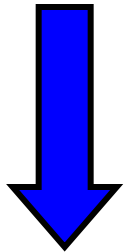
Registers

Dest Value Instruction Done?

F0	<val3>	ADDD F0, F4, F6	Y
F0	<val2>	ADDD F0, F4, F6	Y
F4	M[10]	LD F4, 0 (R3)	Y
--		BNE F2, <...>	N
F2		DIVD F2, F10, F6	N
F10	<val4>	ADDD F10, F4, F0	Y
F0	M[2]	LD F0, 10 (R2)	C

ROB7
ROB6
ROB5
ROB4
ROB3
ROB2
ROB1

Newest



Oldest

F0 = M[2]
F2
F4
F10

To Memory

from Memory

Dest

FP adders

Reservation Stations

Dest

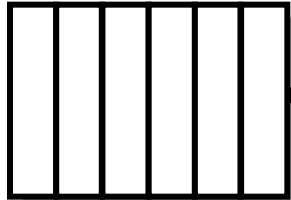
3	DIVD	<val4>, R(F6)

FP multipliers

Dest

Tomasulo With Reorder buffer: 12

FP Op Queue



Reorder Buffer

F0	ROB7
F2	ROB3
F4	ROB5
F10	

RAT

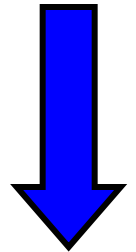
Registers

Dest Value Instruction Done?

F0	<val3>	ADDD F0, F4, F6	Y
F0	<val2>	ADDD F0, F4, F6	Y
F4	M[10]	LD F4, 0 (R3)	Y
--		BNE F2, <...>	N
F2	<val5>	DIVD F2, F10, F6	Y
F10	<val4>	ADDD F10, F4, F0	C
F0	M[2]	LD F0, 10 (R2)	C

ROB7
ROB6
ROB5
ROB4
ROB3
ROB2
ROB1

Newest



Oldest

F0 = M[2]
F2
F4
F10 = <val4>

To Memory

Dest

FP adders

Reservation Stations

Dest

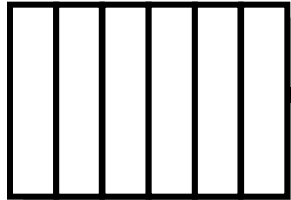
FP multipliers

from Memory

Dest

Tomasulo With Reorder buffer: 13

FP Op Queue



Reorder Buffer

F0	ROB7
F2	
F4	ROB5
F10	

RAT

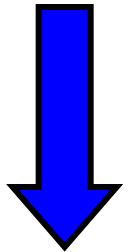
Registers

Dest Value Instruction Done?

F0	<val3>	ADDD F0, F4, F6	Y
F0	<val2>	ADDD F0, F4, F6	Y
F4	M[10]	LD F4, 0 (R3)	Y
--		BNE F2, <...>	N
F2	<val5>	DIVD F2, F10, F6	C
F10	<val4>	ADDD F10, F4, F0	C
F0	M[2]	LD F0, 10 (R2)	C

ROB7
ROB6
ROB5
ROB4
ROB3
ROB2
ROB1

Newest



Oldest

F0 = M[2]
F2 = <val5>
F4
F10 = <val4>

To Memory

Dest

FP adders

Reservation Stations

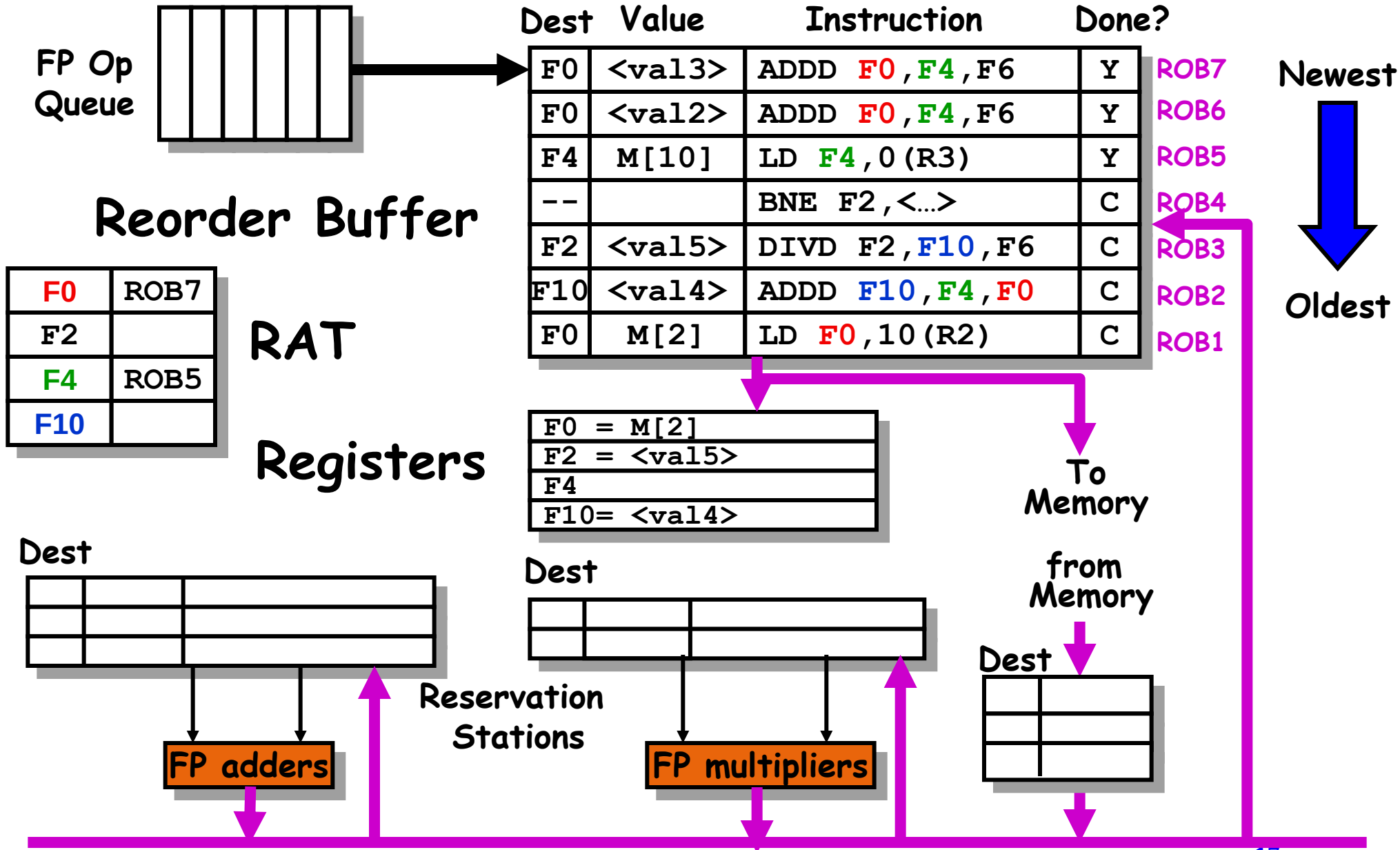
Dest

FP multipliers

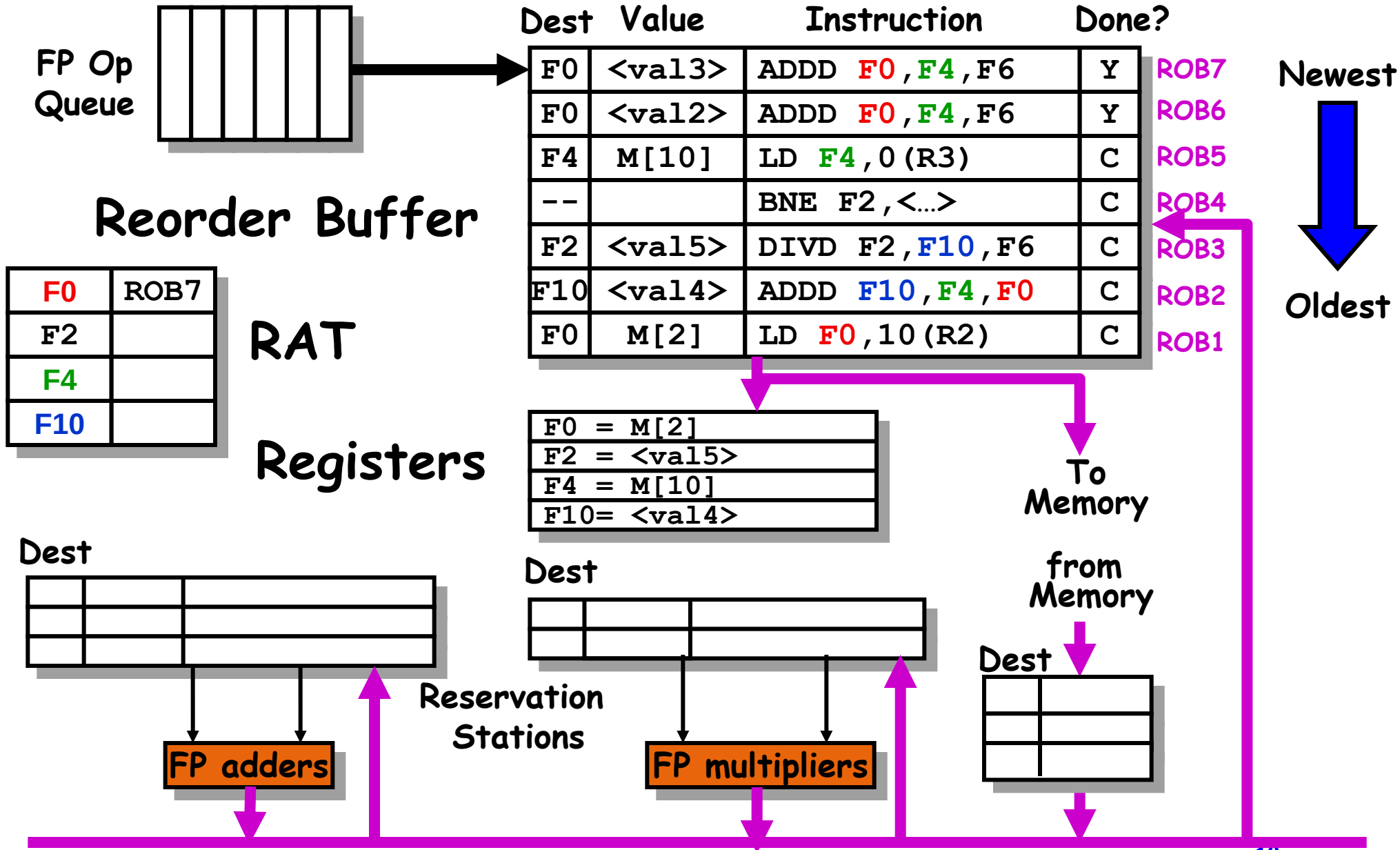
from Memory

Dest

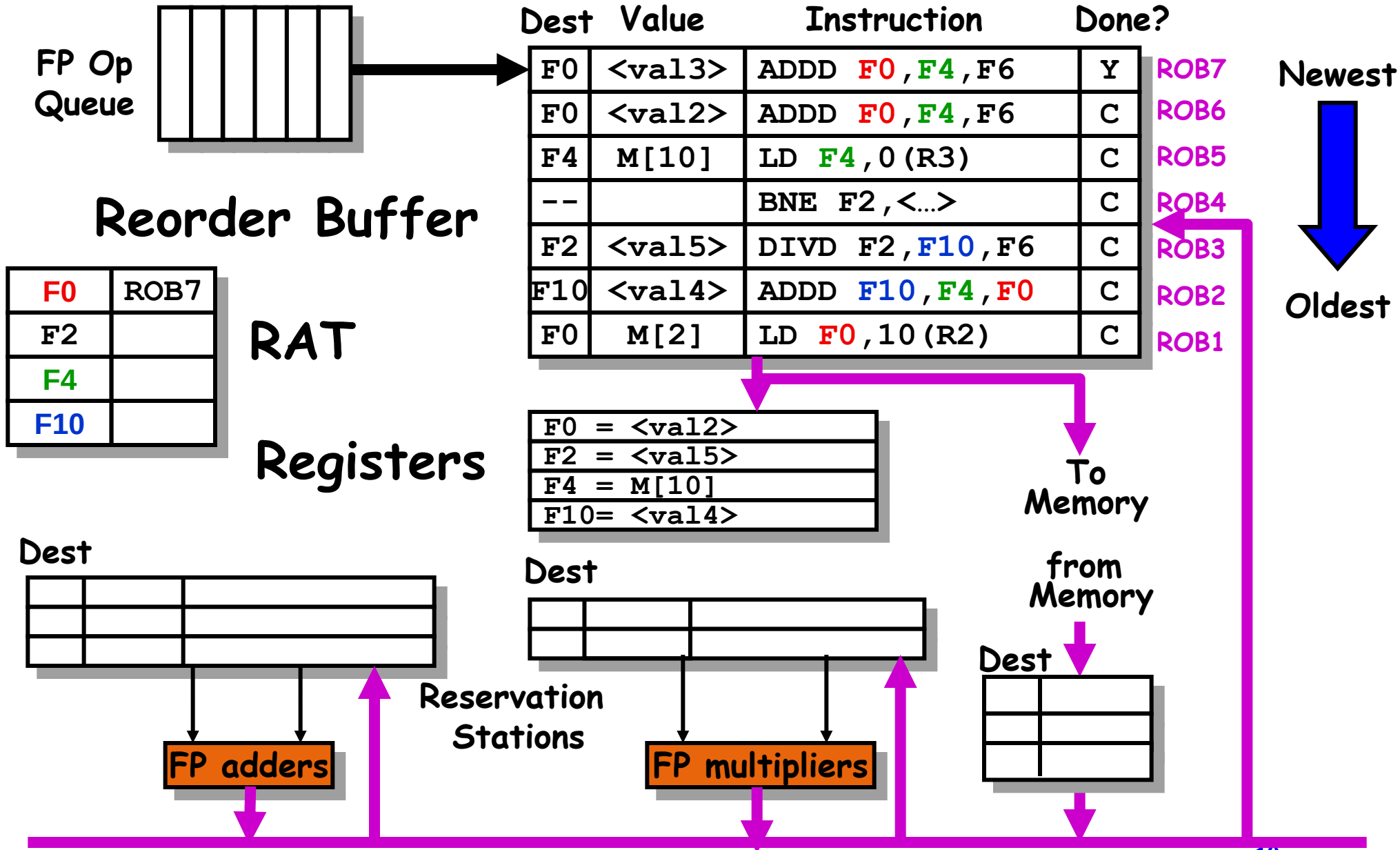
Tomasulo With Reorder buffer: 14



Tomasulo With Reorder buffer: 15



Tomasulo With Reorder buffer: 16



Tomasulo With Reorder buffer: 17

